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Consider an nth order linear d.e with constant coefficients:

$$p_0 \frac{d^n y}{dx^n} + p_1 \frac{d^{n-1} y}{dx^{n-1}} + \cdots + p_{n-1} \frac{dy}{dx} + p_n y = X \quad \rightarrow (*)$$

Here $p_0, p_1, \ldots, p_n$ are constants and X, any function of x

Put $D = \frac{d}{dx}$

Then $(*)$ can be written as

$$(p_0 D^n + p_1 D^{n-1} + \cdots + p_{n-1} D + p_n) y = X \quad \text{— (A)}$$

$A = B$ — (1)
$C = D$ — (2)
(1) + (2) means
$A + C = B + D$

Let $y = y_c$ be a solution of (A) when RHS of (A) is 0

$$(p_0 D^n + p_1 D^{n-1} + p_2 D^{n-2} + \cdots + p_{n-1} D + p_n) y_c = 0 \quad \rightarrow (1)$$

Let $y = y_p$ be the solution of (A)

$$(p_0 D^n + p_1 D^{n-1} + \cdots + p_{n-1} D + p_n) y_p = X \quad \rightarrow (2)$$

Now, (1) + (2) gives:

$$(p_0 D^n + p_1 D^{n-1} + p_2 D^{n-2} + \cdots + p_{n-1} D + p_n)(y_c + y_p) = X$$

which is a solution of (A)

ie $y = y_c + y_p$

is a general solution of (A).

But a gen soln of an 'n'th order d.e will have 'n' arbitrary constants.

Method of soln.

1. First we find y_c with the 'n' arbitrary constants
   We call y_c as the complementary function (C.F)
2. Next we get y_p by solving the given d.e (with no arbitrary constants). This y_p is called as particular integral or P.I.

Getting C.F

Given $(p_0 D^n + p_1 D^{n-1} + \cdots + p_{n-1} D + p_n) y = X$,

the auxiliary equation is:

$$p_0 D^n + p_1 D^{n-1} + \cdots + p_{n-1} D + p_n = 0$$

$[f(D)] \cdot y = X$

For eg: $(2D^3 + 3D^2 + D + 1) y = \sin x$,

the auxiliary eqn is : $f(D) = 0$

$$2D^3 + 3D^2 + D + 1 = 0$$

$ax^2 + bx + c = 0$

$\Rightarrow x = \dfrac{-b \pm \sqrt{b^2 - 4ac}}{2a}$

root

$D^2 - 1 = 0 \Rightarrow D = +1, -1 \rightarrow$ real, distinct

$(D-1)^2 = 0 \Rightarrow D = 1, 1, \rightarrow$ real & repeating

$D^2 + D + 1 = 0 \Rightarrow D = \dfrac{-1 \pm \sqrt{1-4}}{2} = \dfrac{-1 \pm i\sqrt{3}}{2}$, complex

$a=1, b=1, c=1$

root

Finding Complementary fns (C.F)

Let $(p_0 D^n + p_1 D^{n-1} + p_2 D^{n-2} + \cdots + p_{n-1} D + p_n) y = X$

The auxiliary eqn is:

$$p_0 D^n + p_1 D^{n-1} + \cdots + p_{n-1} D + p_n = 0 \quad (*)$$

Case 1: The roots of $(*)$ ie the roots of the auxiliary eqn are real and distinct: $m_1, m_2, \ldots, m_n$

Then the C.F is:

$$y_c = c_1 e^{m_1 x} + c_2 e^{m_2 x} + \cdots + c_n e^{m_n x}$$

[Let the auxiliary eqn be:

$$D - m_1 = 0$$

$\Rightarrow D = m_1$ is the root

ie $\frac{d}{dx} = m_1$

$\Rightarrow \frac{dy}{dx} = m_1 y$

$\Rightarrow \frac{dy}{y} = m_1 dx$

$\Rightarrow \int \frac{dy}{y} = \int m_1 dx$

$\Rightarrow \log y = m_1 x + \log c_1$

$\Rightarrow \log y - \log c_1 = m_1 x$

$\Rightarrow \log_e \left(\frac{y}{c_1}\right) = m_1 x$

$\Rightarrow \frac{y}{c_1} = e^{m_1 x}$

$\Rightarrow \boxed{y = c_1 e^{m_1 x}}$

Case 2: Roots are real but some roots repeat.

- Let $m_1, m_1, m_2, m_3, \ldots, m_{n-1}$ be the roots

  ie the m_1 occurs 2 times

  Then the C.F is:

$$y_c = (c_1 + c_2 x)e^{m_1 x} + c_3 e^{m_2 x} + c_4 e^{m_3 x} + \cdots + c_n e^{m_{n-1} x}$$

$(n-k) = 2$

m_2, m_3 → 1, 2

k m_1, $m_2, \ldots$ $n-k$ → 1

- Let m_1 occur k times, then

$$C.F = y_c = (c_1 + c_2 x + c_3 x^2 + \cdots c_k x^{k-1})e^{m_1 x} + c_{k+1} e^{m_2 x} + c_{k+2} e^{m_3 x} + \cdots + c_n e^{m_{n-k+1} x}$$

Egs

① Find the CF of: $\frac{d^2y}{dx^2} - \frac{dy}{dx} - 6y = 0$

Soln: The auxiliary eqn is:

$$D^2 - D - 6 = 0$$

$\Rightarrow D = 3, -2$ → real & distinct

$3 \to m_1$, $-2 \to m_2$

$\therefore$ C.F is

$$y_c = c_1 e^{3x} + c_2 e^{-2x}$$

② Solve $\frac{d^3y}{dx^3} - 6\frac{d^2y}{dx^2} + 11\frac{dy}{dx} - 6y = 0$

Soln: Aux. eqn is

$$D^3 - 6D^2 + 11D - 6 = 0$$

$\Rightarrow D = 1, 3, 2$

$\therefore$ the C.F

$$y_c = c_1 e^{x} + c_2 e^{3x} + c_3 e^{2x}$$

3. Solve $\dfrac{d^3y}{dx^3} - 5\dfrac{d^2y}{dx^2} + 8\dfrac{dy}{dx} - 4y = 0$